BEATRIZ MENDES ROCHA

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OBJECTIVE

Scientific Computing Engineer dedicated to training and deploying Deep Learning models applied to Computer Vision and Natural Language Processing (NLP).

SKILLS & ABILITIES

Artificial Intelligence: PyTorch, TensorFlow, OpenCV, Transformers (Hugging Face).

Mathematical Modeling: Convex optimization, numerical linear algebra, and multivariable calculus.

MLOps (ML Operations): MLflow, BentoML, Docker, and Kubernetes.

Languages: Python and C++.

EXPERIENCE

Machine Learning Engineer – VisionAI Tecnologia (02/2023 – Present)

- Trained and deployed a convolutional neural network for automated defect detection in industrial parts with **98.4% accuracy**.
- Optimized the inference time of a local Large Language Model (LLM) using quantization techniques, reducing RAM memory consumption by **50%**.

Research Fellow (Undergraduate Research) – UFPR Computer Graphics Laboratory (03/2020 – 12/2022)

- Developed mathematical algorithms for 3D rendering and digital processing of geographic images.

EDUCATION

B.S. in Computational Mathematics – Federal University of Paraná (UFPR) | Completed: 2022

EXTRACURRICULAR ACTIVITIES

Technical Content Creation: Write weekly technical articles on Medium demystifying linear algebra for Deep Learning, accumulating over **5,000 reads**.

Competitive Programming: Participated in the SBC Programming Marathon during undergraduate years, achieving a top 15 regional ranking in solving computational mathematical problems.